

# MAREA: validation design, results to date and the pressure-sensor campaign

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## 1 What is validated, and why

MAREA produces two quantities, and each one is checked separately.

- **The interior tidal lag**  $\tau$  of each 3-km section of the estuary: the delay with which the tide reaches that section after reaching the mouth. Validating it answers two questions: is the lag read from the archive the real one, and does applying it improve the water level assigned inside the estuary?
- **The elevation**  $z$  of each intertidal pixel, from which the emerged time follows. Validating it gives the error of the map and how that error grows inland, where the lag matters most.

Each quantity is tested on site, in the Ría de Villaviciosa, and externally, against records and surveys made by others (Table 1). Nothing that judges a result enters a fit: the inner tide gauges are never used as boundary, and a quarter of the Villaviciosa RTK-GNSS survey was sealed as a test set before any elevation product existed.

Table 1: The four validation blocks.

| block               | reference                                           | sites                                          | metric                                                                  | status                 |
|---------------------|-----------------------------------------------------|------------------------------------------------|-------------------------------------------------------------------------|------------------------|
| lag, on site        | pressure sensors along the ría                      | Villaviciosa                                   | lag in reaching the same level;<br>level error with and without the lag | planned<br>(Section 4) |
| lag, external       | tide-gauge pairs: mouth and<br>inner gauge held out | Ferrol, Westerschelde,<br>Ems-Dollard, Wadden  | same as above                                                           | provisional            |
| elevation, on site  | RTK-GNSS survey                                     | Villaviciosa                                   | RMSE, slope $b$ , $r$                                                   | confirmed              |
| elevation, external | IGN LiDAR (Spain);<br>Vaklodingen bathymetry (NL)   | Ferrol, Villaviciosa;<br>three Dutch estuaries | RMSE, slope $b$ , $r$                                                   | provisional            |

## 2 Metrics

**Lag.** Between a station at the mouth and a station inside, the lag in reaching the same level is the mean, over paired tidal cycles, of  $t_c^{\text{in}} - t_c^{\text{mouth}}$ : the instants at which both records cross the same level on the same limb (rising or falling), at the 25th, 50th and 75th percentile of the mouth’s level. It is compared with the lag MAREA applies at the inner station’s distance from the mouth and with the 95 % bootstrap interval of its section.

**Effect of the lag on the water level.** The level predicted at the inner station is the boundary tide  $h_B$  read  $\tau$  minutes earlier; its error against the recorded level  $h^{\text{in}}$  is

$$\text{RMSE}_h = \sqrt{\frac{1}{N} \sum_t (d(t) - \text{median } d)^2}, \quad d(t) = h_B(t - \tau) - h^{\text{in}}(t), \quad (1)$$

computed with  $\tau = 0$  (the single level every published method assumes) and with MAREA’s lag.

**Elevation.** Product and truth are on different vertical datums, so each series is centred on its own median,  $y_i = z_i - \text{median}(z)$ ,  $x_i = z_i^t - \text{median}(z^t)$ , and the removed offset is reported as the bias. With the truth on the abscissa,

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_i (y_i - x_i)^2}, \quad b = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sum_i (x_i - \bar{x})^2}, \quad r = \frac{\sum_i (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_i (x_i - \bar{x})^2 \sum_i (y_i - \bar{y})^2}}. \quad (2)$$

The slope  $b$  says whether the map reproduces the relief ( $b = 1$ ) or compresses it ( $b < 1$ );  $r$ , how much of it the map explains. Only truth within the range of tide levels the archive sampled ( $\pm 0.25$  m) is scored.

### 3 Results to date

#### 3.1 Elevation on site: RTK-GNSS survey, Ría de Villaviciosa (confirmed)

The survey holds 361 fixed points taken at the lowest spring low water of the month; the development subset is scored here, the test subset stays sealed. Both maps share the same archive, epoch (2023–2025) and boundary (EOT20 at the mouth).

Table 2: Elevation against the RTK-GNSS survey, Villaviciosa development subset.  $n$ : surveyed points the map resolves; RMSE,  $b$ ,  $r$  as in Eq. (2); bias: ellipsoidal heights against the tide-model datum, removed before scoring.

| method                                  | $n$ | RMSE (m) | $b$  | $r$  | bias (m) |
|-----------------------------------------|-----|----------|------|------|----------|
| MAREA                                   | 194 | 0.193    | 0.72 | 0.86 | −52.8    |
| Granadeiro et al. (2021), reimplemented | 163 | 0.275    | 0.39 | 0.63 | −53.3    |

The lag applied on the surveyed pixels is 5 min in median and 16 at most: Villaviciosa tests the elevation estimator more than the lag, which is why the on-site lag campaign of Section 4 is needed.

#### 3.2 Lag against tide-gauge pairs (provisional)

Table 3: Interior lag against tide-gauge pairs, 2023–2025. Lag between the mouth and the inner gauge in reaching the same level (positive: inner gauge later); lag MAREA applies at the inner gauge, with the 95 % bootstrap interval of its section; RMS difference between both gauges at the same instant (the error of one uniform level); level error at the inner gauge from the boundary, Eq. (1), without and with the lag.

| site (inner gauge, km)             | lag, gauges (min) | lag, MAREA (min)  | same instant, RMS (m) | level error (m): no lag → MAREA |
|------------------------------------|-------------------|-------------------|-----------------------|---------------------------------|
| Ferrol (Ferrol2, 10.1)             | +4                | 0 [0, +2]         | 0.10                  | 0.112 → 0.112                   |
| Westerschelde (Terneuzen, 21.9)    | +19               | +7 [+4, +17]      | 0.28                  | 0.356 → 0.311                   |
| Ems-Dollard (Delfzijl, 24.4)       | +51               | +63 [+54, +72]    | 0.50                  | 0.620 → 0.358                   |
| Wadden Sea, Vlie (Harlingen, 18.3) | +88               | +129 [+123, +138] | 0.53                  | 0.562 → 0.303                   |

### 3.3 Elevation against external references (provisional)

Table 4: Elevation against the IGN 5 m LiDAR DTM (PNOA) at the Spanish sites and the Rijkswaterstaat Vaklodigen at the Dutch sites, on all intertidal pixels each map resolves, Eq. (2). The Spanish LiDAR was flown with the flats under water: most of its intertidal cells hold the water surface (share in brackets: cells within 5 cm of the modal value), so it discriminates nothing there.

| method                                               | $n$     | RMSE (m) | $b$  | $r$  |
|------------------------------------------------------|---------|----------|------|------|
| <i>Villaviciosa</i> , IGN LiDAR (69 % water surface) |         |          |      |      |
| MAREA                                                | 7 458   | 0.803    | 0.47 | 0.33 |
| Granadeiro                                           | 4 477   | 0.775    | 0.32 | 0.25 |
| <i>Ferrol</i> , IGN LiDAR (78 % water surface)       |         |          |      |      |
| MAREA                                                | 51 716  | 0.736    | 0.46 | 0.13 |
| Granadeiro                                           | 49 262  | 0.732    | 0.36 | 0.09 |
| <i>Westerschelde</i> , Vaklodigen                    |         |          |      |      |
| MAREA                                                | 62 181  | 0.435    | 0.76 | 0.92 |
| Granadeiro                                           | 45 005  | 0.777    | 0.78 | 0.77 |
| <i>Ems-Dollard</i> , Vaklodigen                      |         |          |      |      |
| MAREA                                                | 412 565 | 0.363    | 0.65 | 0.84 |
| Granadeiro                                           | 98 953  | 0.531    | 0.39 | 0.58 |
| <i>Wadden Sea (Vlie)</i> , Vaklodigen                |         |          |      |      |
| MAREA                                                | 348 894 | 0.329    | 0.81 | 0.68 |
| Granadeiro                                           | 65 576  | 0.477    | 0.65 | 0.44 |

## 4 Further work: the pressure-sensor campaign in Villaviciosa

The Ría de Villaviciosa has no tide gauge inside, so the lag MAREA reads there has never been checked against a measurement. A short campaign of pressure sensors along the ría measures it directly and closes the on-site lag block. The MAREA lags to be tested already exist, so the campaign is a prediction checked afterwards, not a calibration.

**Stations.** One sensor per section of the product, plus the mouth as reference (Table 5). Each is fixed to a stake or mooring at the channel margin, as low as the channel allows, so that it stays submerged through as much of the tidal cycle as possible; where the channel dries at low water, lags are computed only for levels above the sensor.

Table 5: Planned stations and the MAREA lag each one will test (current Villaviciosa product, 2023–2025; sections of 3 km from the mouth).

| station       | position               | from mouth (km) | MAREA lag (min) [95 % interval] |
|---------------|------------------------|-----------------|---------------------------------|
| P0            | mouth, reference       | $\approx 0$ –1  | —                               |
| P1            | section 1 centre       | $\approx 4.3$   | 3.8 [0.0, 6.7]                  |
| P2            | section 2 centre       | $\approx 7.6$   | 20.5 [12.0, 29.6]               |
| P3 (optional) | inner end of the flats | $\approx 9$     | extrapolated from section 2     |

### Instruments and deployment.

- Absolute pressure loggers at every station and a barometer on land (or a logger kept in air) for atmospheric compensation, all sampling at 1 min (5 min at most) and synchronised to UTC before deployment; clock drift is read at recovery and corrected linearly.

- At least 29 days, one full spring–neap cycle, so that every level and both limbs are sampled over the range the archive saw.
- Each sensor’s elevation  $z_s$  surveyed by RTK-GNSS at deployment and at recovery, in the same frame as the elevation survey.
- A conductivity reading (CTD cast or conductivity logger) at each station at deployment and recovery, because brackish water changes the density  $\rho$  and therefore the level scale by up to 2–3 %. This affects the amplitude of the record, not its timing.

**Processing.** Pressure is converted to water level with

$$h = z_s + \frac{p - p_{\text{atm}}}{\rho g}, \quad (3)$$

with  $p$  the logger pressure,  $p_{\text{atm}}$  the barometer record and  $\rho$  from the measured salinity ( $1025 \text{ kg m}^{-3}$  for sea water). Each record is put on a common 1-min grid, spikes removed, gaps longer than one hour left open, and demeaned. The code already exists in the library (module `lag_validation` of `pyintertidal`); the same functions produced Table 3.

**Analysis.** Every station is set against P0 exactly as the gauge pairs of Table 3:

1. the lag in reaching the same level, per limb, at the 25th, 50th and 75th percentile of P0’s level, compared with the MAREA lag of Table 5;
2. the same-instant level difference, the error of one uniform level over the ría;
3. the level error at each station from the boundary, Eq. (1), with  $\tau = 0$  and with MAREA’s lag;
4. at the Sentinel-2 overpasses that fall inside the campaign, the measured local water level against the one MAREA assigns (boundary plus lag): the exact quantity the elevation inversion uses.

**What counts as success, fixed in advance.** The measured lag at each station falls inside the 95 % interval of its section, and applying MAREA’s lag lowers the level error of Eq. (1) at the inner stations. A measured lag outside the interval is reported as such and fed back as a known limitation, not tuned away.

**External elevation references.** Beyond Villaviciosa, the elevation will be tested against bathymetric LiDAR or official bathymetries wherever they cover intertidal ground at low water, to establish the error and its growth inland across regions and tidal regimes. The Dutch Vakkodding already serves this purpose (Table 4); the Spanish national LiDAR does not, since it records the water surface over the flats.